

Things You Wanted to Know!

Q1. I have two questions. First, how does a processor or controller access data from memory? Second, what is Raspi?

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Through email

A1. *Accessing data from memory in a microcontroller (MCU).* Microprocessors and MCUs are different in that, microprocessors do not have inbuilt data memory (RAM) and program memory (ROM), whereas MCUs have both on the chip itself.

In MCUs, operation codes, operands and results are stored in the memory space (ROM and RAM), which is organised into banks segments or pages.

Therefore an arrangement is needed for reaching a location in the data memory space, where the desired operands required by an instruction are stored and where the results after execution of the instruction, if required, are to be stored.

An addressing mode specifies how to calculate the effective memory address of an operand by using the information held in registers and/or constants contained within a machine instruction.

Addressing modes depend on the design of the central processing unit (CPU), memory space and organisation of registers. These differ from MCUs of one brand to another.

There are different ways for addressing a location in the memory space, which are called addressing modes. Some of these are immediate, register direct, absolute, address register indirect, address register relative and relative. In short, data is lodged in data memory space and has address. To get the data, the address is computed by the addressing mechanism.

Let us see how data memory is managed in microcontroller

PIC18F4520. Data memory is organised in 16 banks. However, only six banks have been implemented. Each bank has 256 slots and each slot of memory can accommodate one byte or eight bits. So, the addressing mechanism has to compute the address for one slot out of 1536 slots.

Twelve bits will be needed for this; four bits from the bank selecting resistor (BSR) for selecting the bank. Once we are in that bank, further eight bits will be required for selecting the intended slot out of 256 slots in the selected bank. Each of the 12 bits is made up in a different manner depending upon the addressing mode.

In direct addressing mode, all 12 bits are provided by the instruction.

In indirect addressing mode, the instruction provides the pointer and from that pointer, 12 bits are configured to compute the selection of the bank and the address of the intended location in that bank.

Raspberry Pi. Raspberry Pi, or Raspi, is a computing system that uses a Raspi board, laptop, standard keyboard, mouse, computer monitor or TV, micro USB cable, microSD card with adapter, HDMI cable and network cable for its operations. Programming is done in languages like Python, Wiring Pi, C/C++ and PHP.

It is capable of doing everything that is expected from a desktop computer, from browsing the Internet and playing high-definition video to making spreadsheets and playing games. The system has the ability to interact with the outside world and has been used in many projects, from music machines to weather stations.

Raspi board has 26 dedicated GPIO pins, including a UART, an I2C bus, SPI bus with two chip selects, I2S audio, 3.3V, 5V and ground. The maximum number of GPIOs can theoretically be indefinitely expanded by making use of the I2C or SPI bus.

Interfacing of the camera module is an interesting application of Raspi. It can take high-definition videos as well as still photographs.

Q2. What is the difference between CDMA and GSM technology? Which, according to you, is better?

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A2. Code division multiple access (CDMA) and global system for mobiles (GSM) are two major radio systems used in mobile phones. These incorporate multiple-access technologies, multiple phone calls or Internet connections into one radio channel. CDMA is most popular in the USA, whereas GSM is adopted by almost all other countries as the way to communicate via mobile calls.

Differences between the two are:

GSM comes from an industry consortium and CDMA is owned by chip-maker Qualcomm. It is much easier to swap phones on GSM networks using a SIM card. A carrier must accept any GSM-compliant phone, so GSM carriers do not have total control of the phone you are using.

CDMA carriers use network based white lists to verify their subscribers. Therefore you can only switch phones with your carrier's permission, and a carrier does not have to accept any particular phone onto its network.

3G CDMA networks cannot make voice calls and transmit data at the same time, which is not the case with 3G GSM networks.

GSM is based on time-division system. Voice is transformed into digital data, which is given a channel and a time slot. The receiver listens only to the assigned time slot and pieces the call back together.

CDMA is a code-division system. Every call's data is encoded with a unique key and calls are all transmitted at once. Each receiver has the unique key to divide the combined signal into its individual calls.

In short, GSM or CDMA ultimately provide the same service and the quality of a network depends on the carrier.

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